

12. The earth is nearest to the sun on –

- (a) 3rd January (b) 4th July
(c) 22nd March (d) 21st September

**U.P.P.S.C. (GIC) 2010
Uttarakhand P.C.S. (Mains) 2006**

Ans. (a)

The distance between the sun and the earth is minimum in the first week of January, this situation or position is called 'Perihelion'. Generally Perihelion occurs on 4th January. While opposite to it is on 4th July, the sun stands farthest from the earth and this position or situation is known as "Aphelion".

13. Size of the Sun is how many times bigger than that of the earth ?

- (a) 124 times (b) 100 times
(c) 109 times (d) 115 times

M.P.P.C.S. (Pre) 2014

Ans. (c)

According to NASA, the diameter of the sun is 109.2 times the diameter of the earth. The mean radius of the sun is 695700 km while the mean radius of the earth is 6371 km.

14. Which of the following statements is true regarding earth movement?

1. Earth rotates on its axis from west to east resulting into seasonal changes.
2. Earth revolution around sun results into occurrence of day and night on earth.

- (a) Both 1 and 2 are true (b) Only 2 is true
(c) Both 1 and 2 are false (d) Only 1 is true

Jharkhand P.C.S. (Pre) 2021

Ans. (c)

Night and day occur on the earth due to the rotation of the earth on its axis from west to east, whereas due to the revolution of the earth around the sun, the seasons change on the earth.

15. The bright outer layer of the sun is known as :

- (a) Photosphere (b) Ionosphere
(c) Mesosphere (d) Sunspots

Uttarakhand P.C.S. (Pre) 2024

Ans. (a)

The bright and visible region above the outermost region of the sun 'Convection Zone' is called the photosphere. This is the innermost zone (layer) of the sun's atmosphere. The Temperature of the photosphere 5500-6000°C.

iv. The Mercury

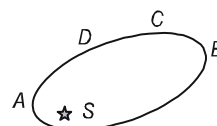
*It is the smallest planet of the solar system and is nearest to the sun. It completes its one revolution in around 88 days.

*Mercury and Venus are the planets having no satellites. *The absence of atmosphere makes it impossible for the sustenance of any life on Mercury. *The average range of temperature on Mercury is around **167°C**. ***Mariner 10** was a satellite launched by NASA that traveled very close to Mercury and took photos of the planet. From the pictures, it was found that the surface of the planet Mercury has a number of mountains and plains. The distance between the Sun and Mercury is approx. 57.9 million km or 0.39 AU.

*All the 8 planets of the solar system revolve around the sun. The time taken for one revolution around the sun, by every planet is as follows – Mercury (88 days), Venus (224.7 days), Earth (365.2 days), Mars (687 days), Jupiter (11.86 years), Saturn (29.4 yrs), Uranus (84.07 yrs), Neptune (164.9 yrs).

***Albedo** is an expression of the ability of surfaces to reflect sunlight (heat from the Sun). *Light coloured surfaces return a large part of the sun's rays back to space (high albedo). Dark surfaces absorb the rays from the sun (low albedo). Hence, Albedo determines the brightness of a body. *Mercury receives a very high amount of sunlight but its albedo is low as compared to earth hence the planet Mercury is not that bright in nature. *The albedo of Earth is 0.3 whereas the albedo of Mercury is mere 0.1. The diameter of planet Mercury is approx. 4879 km.

1. The Planet Mercury is revolving around the Sun in an elliptical orbit, as shown in the given figure. The Kinetic energy of Mercury is greatest at the point labelled.



- (a) A (b) B
(c) C (d) D

I.A.S. (Pre) 1997

Ans. (a)

The closer a planet is to the sun, the stronger the sun's gravitational pull on it. Thus, the planet moves faster.

2. Which of the following planet takes the least time to orbit the sun?

- (a) Pluto (b) Mercury
(c) Earth (d) Saturn

U.P.P.C.S. (Pre) 2002

Ans. (b)